

Project Title: End-to-End Voice Controlled Robotic Manipulation via Vision Language Action (VLA) Models

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1. Problem Statement and Objectives

This project aims to develop an end-to-end embodied AI system that enables a robotic arm to execute physical tasks directly from human voice commands. The system will be built on the Hiwonder SO-ARM101, an open-source 6-axis robotic arm with integrated cameras and compatibility with the LeRobot ecosystem, making it a suitable platform for prototyping embodied AI workflows.

The core problem is the gap between high-level human instruction and low-level robotic action. Recent advances in language models and computer vision allow machines to understand speech and visual scenes, but integrating these capabilities into a single system that can hear a command, interpret its meaning, identify relevant objects, and perform the required motion remains challenging. This is further complicated by the robot's physical constraints, as safely and accurately controlling 6 independent motors is itself a nontrivial problem.

The goal of this project is therefore to design and implement a unified pipeline in which a robot can receive spoken instruction, infer the intended task, perceive the surrounding environment, and use its arm to complete the task autonomously while respecting these physical limitations.

2. Proposed System Design

The system will use a unified VLA framework to create a seamless execution pipeline. First, a speech recognition module will capture the user's voice command and convert the audio into a text transcript. Second, this text instruction will be fed alongside the live camera feed into a pre-trained VLA model. Third, the VLA model will output continuous joint commands based on the text and visual data, considering the physical boundaries of the 6 motors. Finally, these commands will be transmitted to the servos, enabling the robotic arm to execute physical motions safely.

3. Testing Strategy

To ensure reliability, the pipeline will undergo structured testing under varied real-world conditions. Experiments will introduce deliberate uncertainties, including shifted object placements, novel target items, and diverse voice command phrasings. Observing the robot's adaptation to these changes will highlight technical bottlenecks such as visual latency, reasoning errors, or physical instability during 6-motor actuation, enabling iterative refinement toward a robust, practical system.

4. Expected Outcomes and Deliverables

By the conclusion of the research, this project is expected to yield two primary deliverables. First, a technical report detailing the system's responsiveness, processing latency, and robustness against both environmental variations and hardware constraints. Second, a highly reliable robotic agent that demonstrates a consistently high task success rate, with task completion and motor movement accuracy validated through direct human observation.